

MUHAMMAD SAQLAIN

Researcher — ML, CV, Open Source Systems, Educator

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ACHIEVEMENTS

2× Google Summer of Code Selection: Selected for the globally prestigious Google Summer of Code program twice: in 2026 with [NumFOCUS/DeepForest](#) (4.88% acceptance; collaborating with University of Florida researchers on recovering historical forest data for Oregon) and in 2025 with [Google Chromium](#) (5.43% acceptance; earned official saqlain@chromium.org contributor status).

Open Source Mentorship Impact: Mentored underprivileged Pakistani students on competitive open-source programs (GSoC, ESoC, LFX, OSPP) through [Dev Weekends](#). In 2026 alone, 11 students were selected to these brutally competitive programs (9 GSoC, 1 ESoC, 1 LFX). Over 2 years, mentees have collectively earned \$50,138+ in stipends.

Research Associate to Top-Ranked Scientist: Collaborating with the #1 ranked Computer Science researcher from Pakistan (now at PMU, Saudi Arabia) on deep learning CNN techniques for remote sensing.

Stanford Code in Place (2×): Selected twice as a Section Leader for [Stanford's Code in Place](#) (~16,000 enrolled students globally), teaching CS106A course to 26 students from 16 countries over 2 years. [Credential](#)

Competitive Programming: Qualified for [ICPC Asia Topi Regional 2024](#); earned a Bronze Award ranking 252/850+ teams in [CALICO 2024 \(UC Berkeley\)](#); ranked 73rd/500+ in [MIT Informatics Tournament 2025](#); solved 200+ problems on [LeetCode](#) and [Codeforces](#).

English Proficiency: Achieved an overall 7.5 (C1 Advanced) on the [IELTS exam](#).

EDUCATION

University of Education

MS in Information Technology

CGPA: 3.85/4.00

Lahore, Pakistan

August 2025 – 2027

Coursework: Research Methodology, Advanced Information Security, Advanced Topics in Computing, Advanced Computer Networks, Mobile and Wireless Networks, Internet of Thing (IOT)

University of Education

Bachelors in Information Technology

CGPA: 3.13/4.00

Lahore, Pakistan

Dec 2021 – May 2025

Coursework: OOP, DBMS, Algebra, OS, DSA, System and Network Administration, AI, Computer Networks, Discrete Math, Statistics and Probability

RESEARCH EXPERIENCE

GSoC '26 Research Contributor

Weecology Lab, University of Florida – NumFOCUS / DeepForest

Gainesville, FL (Remote)

May 2026 – Present

- Working under 4 research mentors (1 PI, 2 Research Scientists, 1 Research SE) on [Recovering Historical Image Data Using Automated OrthoRegistration](#) (4.88% selection rate).
- Building an automated pipeline to align U.S. Forest Service Aerial Detection Survey (ADS) polygons with year-matched NAIP satellite imagery, correcting 50–500m spatial offsets caused by GPS imprecision and flight geometry errors.
- Developed a hybrid alignment algorithm (ProximityAlign adaptation) combining inside-outside contrast energy, contour-to-boundary distance, and out-of-bounds penalty; improved mean IoU from 0.082 to 0.231 on seed pairs.
- Scaling the pipeline to process ~48,000 Oregon ADS polygons and fine-tuning DeepForest for NAIP-based forest health detection.
- Conducted a [comparative benchmarking study](#) of DeepForest vs DeepTrees, analyzing architectural differences (RetinaNet vs U-Net+watershed), prediction outputs (bounding boxes vs polygons), vegetation indices, and fine-tuning workflows; presented findings to mentors during the selection phase.

Research Associate

Center for Artificial Intelligence (CAI), Prince Muhammad Bin Fahd University

Al Khobar, Saudi Arabia

Jan. 2026 – Present

- Investigating **Computer Vision** architectures for **Remote Sensing** and **Precision Agriculture**, specifically building pipelines for multi-class classification, semantic segmentation, and object detection.
- Developing scalable models for crop disease diagnosis, satellite imaging enhancement, environmental monitoring, and disaster response systems.
- Designing and optimizing neural networks to achieve State-of-the-Art (SOTA) performance on large-scale datasets, integrating emerging advancements in Vision Transformers (ViTs) and hybrid CNN models.

GSoC'25 Chromium OS Contributor

Remote

Google Summer of Code (GSoC) – Web Audio Team

May 2025 – Nov. 2025

- Selected for the Google Summer of Code program (5.43% selection rate) to modernize the testing infrastructure of the Chromium browser engine.
- Engineered a Generative AI automation prototype (Chromium-Automate-CLs) using Gemini models; the tool automated ~90% of test conversion tasks and was highlighted by mentors as a roadmap for future AI-assisted workflows in Chromium.
- Designed a Python heuristic analysis system to parse 326 complex test files based on "Code Complexity Scores" (AST parsing of loops, async logic), creating the data-driven strategy for the migration.
- Delivered 96 CLs (~95% merge rate) to migrate the Web Audio test suite to the W3C `testharness.js` framework, enforcing Google's strict JS style rules and optimizing CI bandwidth usage by ~187 GB/day.

TEACHING EXPERIENCE (volunteer)

Section Leader (2× Selection)

California, United States (Remote)

Stanford University: Code in Place

Apr 2025 – Jun 2025, Apr 2026 – Present

- Selected twice for Stanford's Code in Place, one of the most diverse CS courses globally (~16,000 enrolled students).
- **2026:** Selected from ~2,000 candidates to teach CS106A Python to 11 students from 8 countries: UK (3), Venezuela (2), Serbia, Germany, Peru, Czech Republic, Russia, USA.
- **2025:** Taught Python to ~15 international students from 11 countries: USA (5), Somalia, Thailand, Chile, Philippines, Germany, Ecuador, India, Peru, Canada. Promoted to Section Leader Mentor for 1-on-1 support and taught 6 TeachNow sessions. Credential
- Over 2 years, taught 26 students from 16 countries.

Mentor

Remote

Dev Weekends

May 2024 – Present

- Initiated and led a 6-month open-source mentoring drive, running weekly sessions, proposal reviews, and 1-on-1 guidance for 275+ students on competitive programs (GSoC, ESoC, LFX, OSPP).
- **2026:** 11 students selected (9 GSoC, 1 ESoC, 1 LFX). **2025:** 3 students selected (2 GSoC, 1 OSPP). Collectively, mentees earned over \$50,000 in stipends.
- Delivered 25+ lectures on LeetCode and core DSA, teaching arrays, linked lists, trees, graphs, sorting, searching, and dynamic programming.

Trainer and Moderator

Santa Clara, California Remote

iCodeGuru

Feb 2023 – Present

- Offered volunteer teaching in Python and DSA at iCodeGuru and solved numerous LeetCode problems. All recording are here
- Taught courses on the IELTS speaking and writing modules specifically.

PROJECTS

Your JobBuddy (Final Year Project) GitHub — Sept 2024 – May 2025

- Developed a **GenAI-powered resume optimization platform** using Next.js (TypeScript) to maximize ATS (Applicant Tracking System) compatibility for job seekers.
- Integrated LLMs to implement resume parsing, automated scoring, and context-aware feedback on clarity and formatting.
- Designed a responsive interface featuring real-time document editing, multi-template support, and PDF generation.

QuillPDF Live Demo

- Engineered a **RAG-based (Retrieval-Augmented Generation)** SaaS using LangChain, Pinecone, and OpenAI to enable AI conversations with PDF documents.
- Built a high-performance full-stack architecture with Next.js, Prisma, and Tailwind CSS, supporting free/pro subscription tiers.

ClearOUT (AI Hackathon) Live — Submission

- Led a 6-member international team in the lablab.ai LangFlow Hackathon to build an AI-powered recycling/reuse assistant using image recognition and LangChain/LangFlow.

Full-Stack E-Commerce Platforms Shopo Demo — eCommerc3 Demo

- Built 2 production-grade e-commerce applications using the MERN stack (MongoDB, Express, React, Node.js), implementing Stripe/PayPal payments, role-based dashboards, and 100+ functional modules across both projects.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++

ML & Deep Learning: PyTorch, scikit-learn, NumPy, SciPy, Pandas, OpenCV

Computer Vision & Remote Sensing: Object Detection, Image Registration, GeoPandas, Rasterio, GDAL, QGIS

Web Dev: HTML5, CSS3, React JS, Redux Toolkit, Next.js, Node JS, Express JS

DevOps & Tools: Docker, CI/CD, GitHub Actions, AWS (RDS, S3), Git

AI: Generative AI, LLMs, RAG (LangChain, Pinecone)

Databases: MongoDB, MySQL, PostgreSQL

REFERENCES

Dr. Ethan White — Professor & PI, Weecology Lab @ University of Florida — [LinkedIn](#)

- **Email:** ethanwhite@ufl.edu
- **Role:** GSoC 2026 Mentor (NumFOCUS / DeepForest)
- **Location:** Gainesville, FL, United States

Dr. Ben Weinstein — Research Scientist, Weecology Lab @ University of Florida — [LinkedIn](#)

- **Email:** ben.weinstein@weecology.org
- **Role:** GSoC 2026 Mentor (NumFOCUS / DeepForest)
- **Location:** Gainesville, FL, United States

Hongchan Choi — Tech Lead & Manager, Google (Chromium) — [LinkedIn](#)

- **Email:** hongchan@google.com
- **Web Audio / Web MIDI API Owner:** Chromium Project
- **Adjunct Professor:** Stanford University
- **Location:** Mountain View, CA

Dr. Muhammad Attique Khan — Assistant Research Professor, Center of AI @ PMU — [LinkedIn](#)

- **Email:** mkhan3@pmu.edu.sa
- **Location:** Saudi Arabia